AUTOMOTIVE CYBERSECURITY

WireLock

The $2 Zero-RF Automotive Security Solution.

Eliminating relay attacks with physical cryptographic authentication.



JEYARAM B · +91 8309631052 · jeyaram.reddy.ece@gmail.com



01 · THE EXPOSURE

Keyless entry has a wireless attack surface

- Relay and skimming attacks increasingly compromise keyless entry systems.
- UWB and biometrics add significant hardware cost, often $20+, while still relying on wireless channels.
- Commercial fleets, rental companies, and vehicle owners absorb the resulting theft and disruption.

02 · THE MECHANISM

WireLock Architecture

Move authentication from the air to the physical connection.

- **Zero-RF architecture:** removes the wireless attack surface; no radio frequencies means no relay attacks.
- **Physical handshake:** proprietary contact-based authentication.
- **Advanced cryptography:** ASCON and AES-based device authentication.
- **Dynamic defense:** dynamic block permutation helps prevent hardware-level cloning.



03 · THE FIT

Security that fits the vehicle economics

A low-cost security layer for platforms where premium hardware is out of reach.

Ultra-low cost

Target BOM under $2 at volume production.

Seamless integration

Standalone master key or high-security backup.

Commercial fleets

Protect shared vehicles without premium hardware.

Rental, EV, and 2-wheelers

Built for self-drive rental cars, cost-sensitive EV platforms, and 2-wheelers.

04 · PARTNER VALIDATION

Traction is ready for partner validation

PATENT PENDING

IP protection

Indian patent filed

App No: 202641101016

Prototype

Fully functional hardware prototype with 3D-printed enclosure ready.

Tier-1 automotive partners · technical evaluation · co-development · non-exclusive licensing